

LangChain (Embedding + Vector Database)

```
from dotenv import load_dotenv

from langchain_openai import OpenAIEmbeddings

from langchain_community.vectorstores import FAISS

from langchain_core.tools import create_retriever_tool

load_dotenv()

embedding = OpenAIEmbeddings(model="text-embedding-3-large")

texts = [

    "Apple makes very good computer.",

    "I believe Apple is innovative.",

    "I am a fan of Argentina."

]

vector_store = FAISS.from_texts(texts, embdding = embedding)

print(vector_store.similarity_search("Apples are my favorite fruit.", k = 7))

print(vector_store.similarity_search("Linux is a great OS.", k = 7))

retriever = vector_store.as_retriever(search_kwargs={"k": 3})

retriever_tool = create_retriever_tool(retriever, name="kb_search", description =
"Search the small product" )

agent = create_agent(

    model = "gpt-4.1-mini",
```

```
tools = [retriever_tool],

system_prompt= (

    "You are a helpful assistant. For questions about Macs, Apple, or laptop,"

    "First call the kb_search tool to retrieve context, then answer succinctly. Maybe you  

    have to use it multiple times before answering")

)

result = agent.invoke({

    "message": [{"role": "user",

    "content": "What three fruits does the person likes and dislike?"}]

})

print(result)

print(result["messages"][-1].content)
```